Combinations vs. Permutations: Subsets & Prime Cancellation

Probability, Random Variables & Combinatorics • Standard: CCSS.MATH.HSS.CP.B.9

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Calculate permutations and combinations by evaluating factorials across the 4-register stack and canceling prime factors.

Combinations nCr = n! / (r!(n-r)!) count unordered selections by dividing ordered permutations by duplicate arrangements. On the stack, you can either evaluate and cancel the factorial branches step-by-step across Y and X or verify directly with [nCr].

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Fraction Studio (84 mm Probability Towers) & Prime Factor Studio

Hands-On Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 8 × 7 = 56.

Student Prediction (Before Calculator): How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = 40320 / (6 × 120) = 56.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

4. STORY CONTEXT (Sir Francis Galton's 1873 Bean Machine, The Asteroid Ceres & The Guinness Brewery Chemist)

In 1873, Sir Francis Galton built a board studded with pins (the Quincunx) to show how independent random bounces settle into a bell-shaped curve—while in 1908 at the Guinness Brewery in Dublin, chemist William Sealy Gosset ('Student') invented small-sample statistics with n - 1 degrees of freedom! Students model exact probability ratios on 84 mm Fraction Studio towers, cancel factorial primes, and accumulate trials with [Σ+].

Core Investigation (Student Warm-Up A):

A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Statistical Accumulation: Mean & Variance with [Σ+]

One-Variable Data Analysis & Summations • Standard: CCSS.MATH.HSS.ID.A.2

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Statistical measures of center (Mean x-bar) and spread (Standard Deviation s) summarize an entire distribution of values. Pressing [Σ+] on the stack calculator accumulates n, ⊠x, and ⊠x² simultaneously in dedicated statistical registers so x-bar and s can be read without re-entering data.

Blocks & Prediction: Lay out the 5 sample values (4, 6, 7, 8, 10) with Digit Blocks on the Place Value Tray and accumulate them with [Σ+]. Prediction: Sum = 4 + 6 + 7 + 8 + 10 = 35. Sample size n = 5. Mean x-bar = 35 / 5 = 7.0.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

`4` `Σ+` `6` `Σ+` `7` `Σ+` `8` `Σ+` `1` `0` `Σ+`

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
`4` `Σ+`	1	First score entered: sample size n = 1
`6` `Σ+`	2	Second score: n = 2
`7` `Σ+`	3	Third score: n = 3
`8` `Σ+`	4	Fourth score: n = 4
`1` `0` `Σ+`	5	Fifth score: n = 5 total data points

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Hands-On Block Remediation: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows n > 1, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by (n − r)! when computing C(n, r) = n! / (r!(n − r)!).

Hands-On Block Remediation: Compute n! into Y, compute r! and (n − r)! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

4. EXTENSION MATHEMATICAL INSIGHT

Statistical measures of center (Mean x-bar) and spread (Standard Deviation s) summarize an entire distribution of values. Pressing [Σ+] on the stack calculator accumulates n, ⊠x, and ⊠x² simultaneously in dedicated statistical registers so x-bar and s can be read without re-entering data.

Combinatorics, Probability Ratios & Series Accumulation

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: COMBINATIONS, PERMUTATIONS & STACK FACTORIALS

- Factorial [f:] [x!] multiplies n × (n-1) × ... × 1; Prime Factor blocks show how common prime factors cancel in n! / r!.
- Permutations (order matters): nPr = n! / (n - r)!.
- Combinations (subsets/groups): nCr = n! / (r! × (n - r)!).

Key Sequence: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Investigation 1: Calculate 6! (6 factorial) on the calculator.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Calculate 7C2 (combinations of 2 from 7).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Accumulate 5 and 15 with [Σ+]. What is the mean?

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 11: Combinatorics, Probability Ratios & Series Accumulation — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 11–12 / AP

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1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Core Solution: 210 combinations. Keystrokes: [1][0] [f:][x!] [4] [f:][x!] [÷] [6] [f:][x!] [÷]

Core Check Keys: `1` `0` `↶` `x!` `4` `↶` `x!` `÷` `6` `↶` `x!` `÷`

Extension Challenge (Warm-Up B): Accumulate data points 12, 15, 18, 21 using [Σ+]. What is the sample count n and mean?

Extension Solution: Count n = 4, Mean x-bar = 16.5.

Ext. Check Keys: `1` `2` `Σ+` `1` `5` `Σ+` `1` `8` `Σ+` `2` `1` `Σ+`

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Calculate 6! (6 factorial) on the calculator.

Expected Answer: 720

SC-32 Verification Keys: `6` `↶` `x!`

Investigation 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!).

Expected Answer: 60

SC-32 Verification Keys: `5` `↶` `x!` `2` `↶` `x!` `÷`

Investigation 3: Calculate 7C2 (combinations of 2 from 7).

Expected Answer: 21

SC-32 Verification Keys: `7` `↶` `x!` `2` `↶` `x!` `÷` `5` `↶` `x!` `÷`

Investigation 4: Accumulate 5 and 15 with [Σ+]. What is the mean?

Expected Answer: Mean = 10

SC-32 Verification Keys: `5` `Σ+` `1` `5` `Σ+` `↷` `x̄, ȳ`

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.